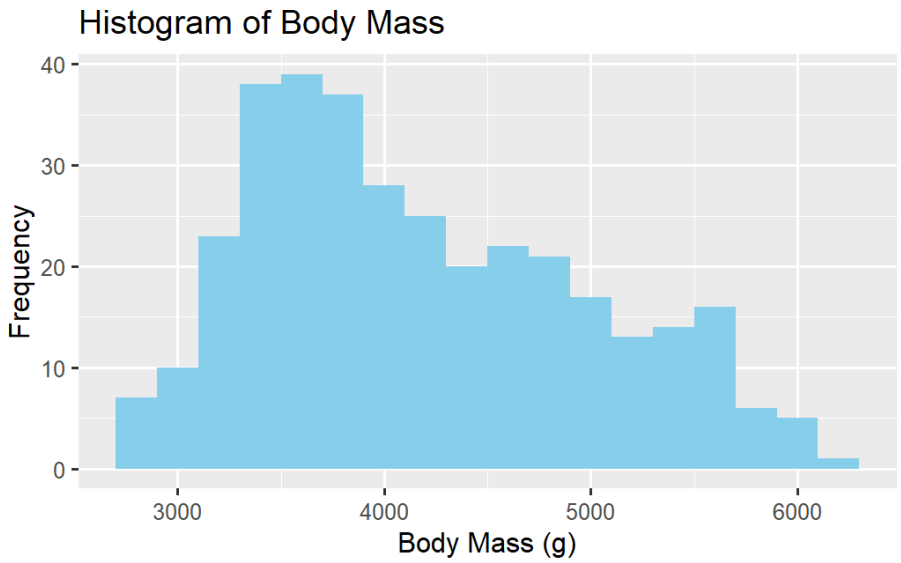
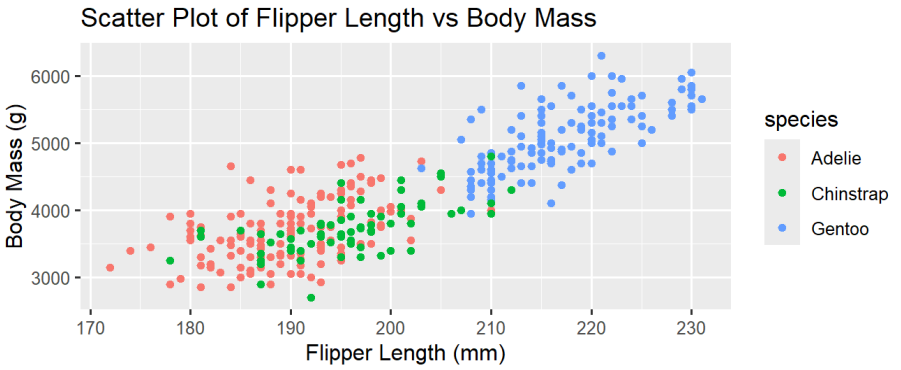
DSCI401 - Homework 6

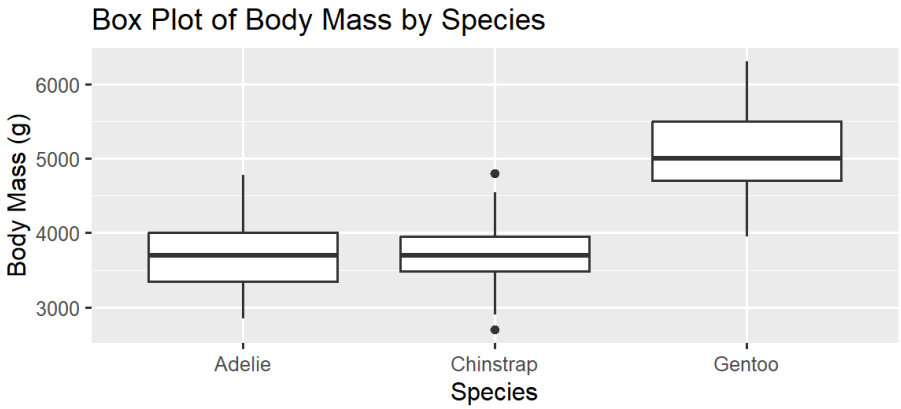
Due: Nov 24, 2026

Homework should be turned in on Sakai.

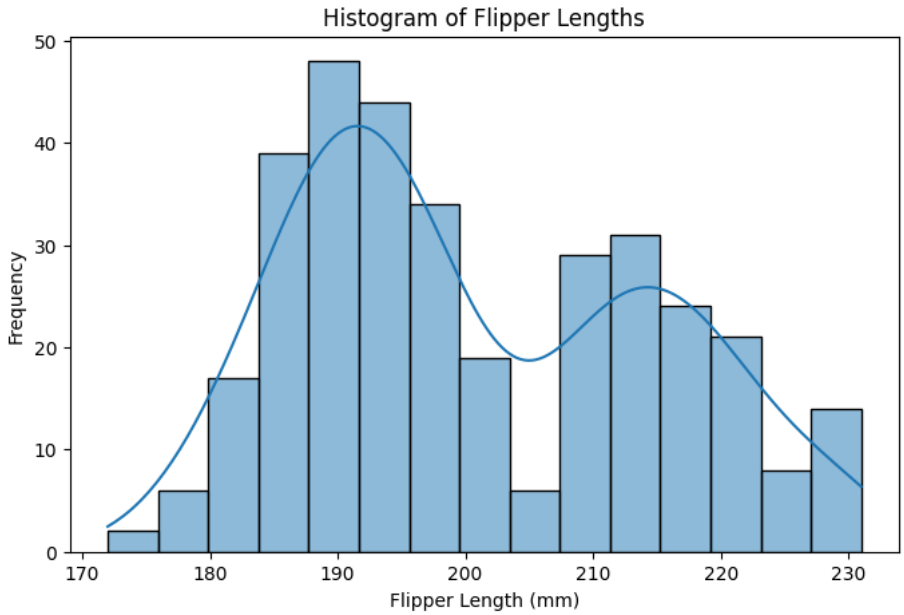
1. (25 points) Create the following images in R:
   Use 'penguins' dataset from the library(palmerpenguins).

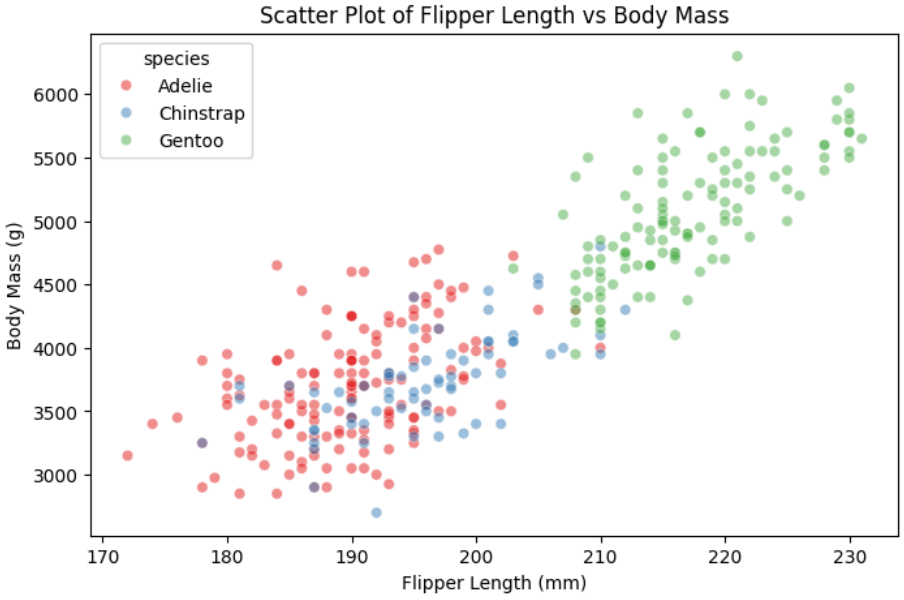


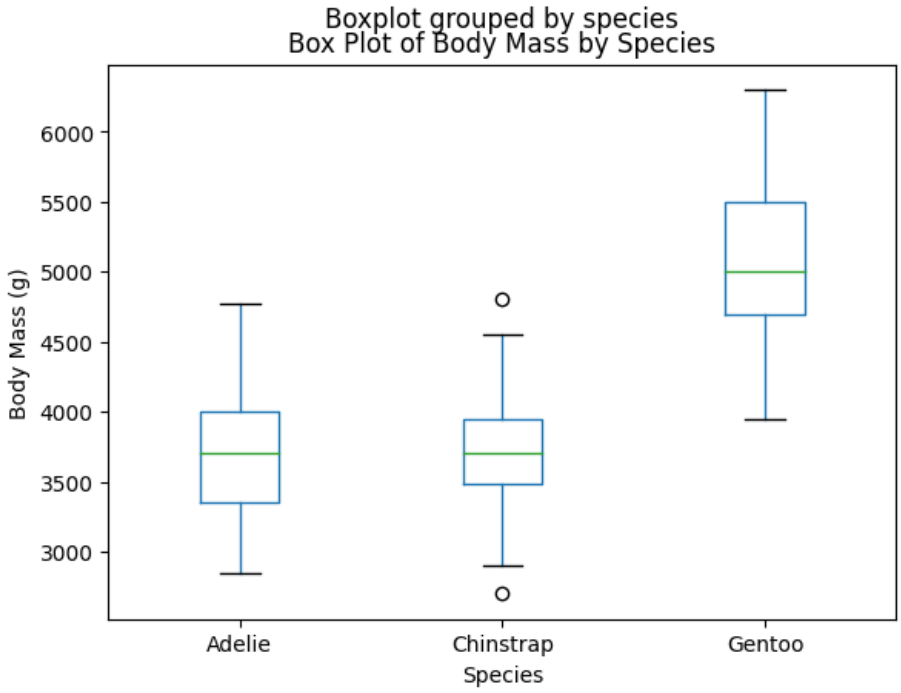




2. (25 points) Create the following images in Python:







3. (10 points) Make the ugliest most stylized figure you can make using the penguins dataset in both R and Python.

4. (40 points) Use the CTA datasets provided (`CTA_ridership.csv` and

`CTA_stops_long.csv`) to explore relationships between ridership and train lines in R and Python.

(a) To combine together both rides and stops datasets together, which variable(s) should you use and why? What would be the benefit or drawback of using these variables? Would this exclude any data from one dataset to the other?

(b) Once joined calculate the total riders per line.

(c) Visualize 2 bar chart i) stations per line ii) riders total per line and match colors of line to color of bar chart (use two purple colors).

(d) Could there be an issue with the way the data was collected from ridership and joining with the stop data to calculate total rides per line? Discuss.